

Pronob Kumar Barman

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PROFESSIONAL SUMMARY

PhD Candidate (Information Systems) and Lead Machine Learning Engineer with 7+ years of experience spanning Embedded AI, FinTech, and Healthcare. Specialized in deploying fairness-aware models and low-latency recommendation systems on ARM-based edge devices and cloud-native (AWS) environments. Proven track record of translating research into production-grade systems under strict regulatory, memory, and latency constraints.

CORE SKILLS

Machine Learning and AI: LLMs, Generative AI, RAG, Reinforcement Learning (PPO), Recommendation Systems, Anomaly Detection, Fairness-Aware ML, Causal Inference, Explainable AI (SHAP, LIME)

Embedded and Edge AI: ARM Cortex-M55 and Cortex-A53, ARM Ethos-U NPU (U65, U85), Vela Compiler, TensorFlow Lite Micro, Quantization-Aware Training, Structured Pruning, ONNX

MLOps and Infrastructure: AWS SageMaker, EKS, Docker, Kubernetes, MLflow, Airflow, CI/CD (GitHub Actions), FastAPI, Terraform, Pinecone, FAISS, Model Monitoring

Programming: Python (PyTorch, TensorFlow, scikit-learn), C/C++, R, Bash, Java, SQL, Spark, Kafka

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (PhD Intern) – Behavioral Health AI

Dec 2025 – Present

Technuf LLC

Rockville, MD

- Increased session retention by 15% and user engagement by 10% by designing and deploying a PPO-based reinforcement learning recommendation engine.
- Architected a cloud-native MLOps platform on AWS SageMaker and EKS, reducing model deployment cycles by 40%.
- Delivered real-time personalized inference with sub-350ms latency via FastAPI microservices for high-concurrency workloads.
- Built HIPAA- and GDPR-compliant ETL pipelines processing over 1M behavioral health records daily.
- Operationalized fairness auditing using SHAP and Fairlearn, reducing demographic parity gaps by 22%.

Machine Learning Engineer (PhD Intern) – Embedded Cybersecurity

Jul 2025 – Nov 2025

Axiado Corporation

San Jose, CA

- Reduced edge-device inference latency by 15% on ARM Ethos-U65 and Ethos-U85 NPUs under strict memory and power constraints.
- Compressed model size by 4x using INT8 quantization and structured pruning, optimized via Vela compiler for Ethos-U deployment.
- Improved early-stage threat detection accuracy by 20% through lightweight transformer-based temporal modeling.
- Built automated embedded ML pipelines with ONNX export, ensuring 99.9% parity between cloud and edge models.

Machine Learning Engineer (Intern) – Generative AI

Jun 2024 – Aug 2024

Technuf LLC

Rockville, MD

- Accelerated security incident response by 40% using retrieval-augmented generation (RAG) pipelines for threat intelligence.
- Reduced low-value alerts by 30% through hybrid threat-scoring models, improving analyst efficiency.

Graduate Research and Teaching Assistant – Healthcare AI

Jan 2022 – May 2024

University of Maryland, Baltimore County

Baltimore, MD

- Developed the PL-NCF (Pseudo-Label Neural Collaborative Filtering) framework to address extreme interaction sparsity in Online Health Community (OHC) recommendations, achieving a 2x increase in HR@5 (from 2.65% to 5.30%) using dual-representation learning.
- Discovered and quantified the “separability-accuracy paradox,” identifying a significant negative Spearman correlation ($\rho \approx -0.59$) between ranking performance and the clusterability of main-branch embeddings.
- Architected the gDMM and gSTM probabilistic topic models to automate personalized support group formation by integrating user-generated text, demographic metadata, and Node2Vec network embeddings, achieving +15% coherence score.

and 2x improvement in topic distinctiveness.

- Evaluated large-scale OHC datasets (2M+ users, 8M+ posts), validating that group-specific models consistently outperform standard LDA, DMR, and STM baselines in semantic coherence and predictive accuracy.
- Conducted qualitative thematic analysis with a second-coder reliability of 0.78 (Cohen's Kappa), validating that model-generated groups align with chronic illness management and mental health themes.
- Mentored 100+ students in Data Mining and Cybersecurity Analytics; received Excellent TA Award with 80% of students providing 5-star reviews.

Deputy Director – AI Systems and Data Science Leadership

Apr 2018 – Dec 2021

Bangladesh Bank

Dhaka, Bangladesh

- Led national AI modernization initiatives deployed across 12 regional branches, integrating ML into financial auditing and policy analytics and reducing manual audit hours by 5%.
- Designed AML systems using XGBoost with rule-based flagging, processing \$1M+ in transactions annually and identifying over 200 high-risk cases via anomaly detection.
- Developed a Bengali-language NLP chatbot, reducing central bank support ticket volume by 25%.
- Built time-series forecasting models improving inflation projection accuracy by 18%.

EDUCATION

PhD in Information Systems (AI and Machine Learning)

Expected May 2026

University of Maryland, Baltimore County

MS in Statistics

May 2021

University of Kentucky

MS in Statistics

Jan 2017

University of Dhaka

BS in Statistics

Sep 2015

University of Dhaka

SELECTED PUBLICATIONS

- Barman, P. K., Reynolds, T. L., and Foulds, J. Facilitating online healthcare support group formation using topic modeling. *Studies in Health Technology and Informatics*, 2025.
- Balogun, Z. A., Barman, P. K., Onwumbiko, B. K., and Reynolds, T. L. User experiences with mobile personal health records. *AMIA Annual Symposium Proceedings*, 2024.

AWARDS

Excellent Teaching Assistant Award, University of Maryland, Baltimore County

2023

UMBC Graduate Student Association Research Grant (\$1,000 Award)

2024

NSF Travel Grant, SIAM SDM Doctoral Forum

2025